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Interpretable temporal-spatial graph attention network for multi-site PV power forecasting

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ABSTRACT

Accurate forecasting of photovoltaic (PV) and wind production is crucial for the integration of more renewable energy sources into the power grid. To address the limited resolution and costs of methods based on numerical weather predictions (NWP), we take PV production data as main input for forecasting. Since PV power is affected by weather and cloud dynamics, we model spatio-temporal correlations between production data by representing PV systems as nodes of a dynamic graph and embedding production data, geographical information and clear-sky irradiance as signals on that graph. We introduce a new temporal-spatial multi-windows graph attention network (TSM-GAT) for predicting future PV power production. TSM-GAT can adapt to the dynamics of the problem, by learning different graphs over time. It consists of temporal attention with an overlapping-window mechanism that finds the temporal correlations and spatial attention with a multi-window mechanism, which captures different dynamical spatio-temporal correlations for different parts of the forecasting horizon. Thus, it is possible to interpret which PV stations have the most influence when making a prediction for short-, medium- and long-term intra-day forecasts. TSM-GAT outperforms multi-site state-of-the-art models for four to six hours ahead predictions, with average NRMSE 12.4% and 10.5% on a real and synthetic dataset, respectively. Furthermore, it outperforms state-of-the-art models that use NWP as inputs for up to five hours ahead predictions. TSM-GAT yields predicted signals with a closer shape to ground truth than state-of-the-art models, which indicates that it is better at capturing cloud motion and may lead to better generalization capabilities.

1. Introduction

It is difficult to integrate variable solar power sources into the existing power grid due to the dependence of their production on weather conditions, including irradiance and cloud motion. In particular, intermittent and stochastic characteristics of photovoltaic (PV) power represents a challenge for energy trading and grid congestion management [1]. Thus, it is crucial to accurately forecast PV power generation.

All deterministic PV forecasting models aim at increasing the accuracy of the intra-day (up to six hours ahead) forecast use inputs from different sources, including ground-based cameras, satellite images, and numerical weather predictions (NWP); the interested reader may check the work of Antonanzas et al. [2] for in-depth review. Although ground-based cameras yield high accuracy for short-term forecasts, they can be expensive to deploy and maintain. On the other hand, satellite based-images and NWP [3–5] excel at long-term (six-hours to day ahead) forecasts, but perform rather poorly for short-term and high spatial resolution forecasts. Hence, in order to avoid the aforementioned issues from the additional data sources, recent works use only the past PV data from multiple sites to exploit the spatio-temporal relations and improve the forecast accuracy. They follow the intuition that, in a dense network, PV stations act as virtual weather stations, and look at the problem as a multi-site PV forecasting problem.

PV production is highly intermittent, due to the stochastic cloud behavior and cloud dynamics. Thus, in order to obtain an accurate prediction, a model that takes into account dynamically changing spatio-temporal correlations between the production of different PV sites is required. Various machine learning models have been used in order to address this issue. The traditional linear auto-regressive methods in the works of Cavalcante and Bessa [6], Agoua et al. [7], Singh and Pozo [8] and Carrillo et al. [9] can, however, not capture

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non-linear temporal correlations. Thus, neural networks are utilized to model spatial and temporal correlations and they include recurrent and convolutional neural networks [5,10–12]. The work of Jeong and Kim [12], reorders signals based on the locations, creating a fixed structure, similarly to the work of Zhu et al. [13] and then uses convolutional neural networks. Attention mechanisms that have recently increased accuracy in sequence prediction tasks have been introduced in the PV power forecasting problems, coupled with recurrent and convolutional structures [10,14,15]. The models presented in the works of Lai et al. [10] and Shih et al. [15] use only a limited number of data steps from previous days as input to the model, thus, neglecting temporal shift and periodicity.

In order to fully exploit spatio-temporal correlations between sites and infer part of the cloud dynamics, the PV power forecasting task has also been addressed from a Graph Signal Processing (GSP) perspective [16]. It uses PV power production at certain time steps in auto-encoder graph convolutional structures, modeling only spatial relations when making a prediction. Both spatial and temporal relations are modeled in the deterministic PV forecasting, where graph convolutional structures are applied [17], yielding the encoder–decoder Graph convolutional long-short term memory and Graph convolutional transformer solutions. Although these advanced architectures capture spatio-temporal correlations, they are restricted to predefined k-neighbors graphs, neglecting the impact of the nodes which are further away, which is important for longer-term (from 2 to 6 h ahead) forecasts. Furthermore, when modeling PV power production, we model physical phenomena, thus, an interpretable model is desirable. On the one hand, recurrent and convolutional models are difficult to interpret. On the other hand, state-of-the-art attention-based architectures for time series forecasting use multi-head attention which is not fully interpretable. The works of Michel et al. [18] and Cordonnier et al. [19] have shown that the multi-head attention tends to learn redundant relationships. Moreover, the work of Baan et al. [20] demonstrated that the multi-head approach is only partially interpretable and concluded that in order to be transparent it should have no overlap in specialization, but focus on distinct representational subspaces. Therefore, an attention-based model which focuses on distinct representational subspaces is required in order to be able to understand what influences the forecast.

Multi-site time series forecasting task has been studied not only in the PV domain but also in wind speed, wind power production, electricity and irradiance forecasting. In the work of Ghaderi et al. [21] recurrent long-short term memory (LSTM) cells are used to extract temporal correlations. In addition to recurrent neural networks, convolutional neural networks were used in the work of Lai et al. [10]. However, they were outperformed by the attention mechanisms, which capture correlations due to their ability to select relevant time steps and locations; see the work of Shih et al. [15]. Graph convolutional structures coupled with recurrent structures proposed in wind forecasting task in work of Khodayar and Wang [22], requires a different model for each step ahead prediction, which is not scalable. The lower scalability and higher error are some of the shortcomings which have been addressed by introducing the graph multi-head attention. However, graph multi-head attention is not fully interpretable, as previously discussed. All these spatio-temporal networks restrict the graph topology to a small number of k neighbors.

On the other hand, a lot of work has been focused on single-site forecasting task for PV and wind power production. The models proposed in the works of Azad et al. [23] and Hossain et al. [24] focus on the long-term (up to year ahead) forecasts of wind speed and power density. However, these approaches require learning different models for each station which increases the computational costs. Recent single-site PV power production models in the works of Zhou et al. [14], Kharlova et al. [25], Ren et al. [26] and Dairi et al. [27] use LSTM to find temporal dynamics and attention mechanism to extract relevant features. They show that LSTM coupled with the attention mechanism improves accuracy, especially in the case of long sequence inputs. However, these methods also exploit NWP and weather data, which have low spatial and temporal resolution for multi-site forecasting models with high number of PV plants. Moreover, by taking into account only single-site time series, it is not possible for these models to exploit spatial correlations between the sites. The addition of LSTM cell to the attention block makes it more difficult to interpret on which temporal steps the model is focusing when making the prediction.

In this work, we propose a temporal-spatial multi-window graph attention network (TSM-GAT) applied to the PV power forecasting problem to solve the aforementioned issues of low interpretability and restricting graph topology to a small number of the closest neighbors, thus excluding the long-term correlations between sites. A dense network of PV stations could be used as a network of virtual weather stations, where spatio-temporal correlations are modeled using graphs. Thus, PV stations are modeled as nodes in a dynamic graph, and embedded past PV data, coupled with geographical information and clear-sky data, represents the graph signals. TSM-GAT consists of temporal attention with an overlapping-window mechanism and spatial attention with multi-window mechanism. The output of the temporal attention is temporal correlations and embedded temporal features which are passed to the spatial attention, which finds different spatio-temporal correlations. Furthermore, in the spatial attention, a multi-window mechanism is introduced in order to increase interpretability and accuracy at longer-term horizons. Moreover, this model yields the shape of the predicted signal closer to the shape of the ground truth, which indicates that the model is better at capturing the movement of clouds and cloud coverage. Since PV production is dependent on weather conditions and information from PV stations is limited to certain spatial location, PV stations across different countries have different weather conditions. The interpretability of the model indicates a close link between the model and physical phenomena, making it promising in terms of generalization and performance in different weather conditions. The contributions of the current work include:

- A TSM-GAT model that extends of graph attention networks for time series forecasting tasks. Temporal attention embeds temporal features from times-series data into temporal windows, creating inputs for the dynamic graph. The following spatial attention finds spatio-temporal correlations for each node and each temporal window, yielding a dynamical adjacency.
- A multi-window mechanism is developed in TSM-GAT to allow the model to learn different dynamical adjacency matrices for shorter-, medium- and longer-term predictions. Thus, it is possible to interpret which nodes the model focuses on for different forecasting horizons.
- A performance evaluation of TSM-GAT on real and synthetic PV datasets and comparison with state-of-the-art multi-site and single-site models. For multi-site models, an analysis of the impact of weather conditions, and node density on the forecast is presented on the real dataset in Switzerland, in order to understand what drives the performance. A study has further been conducted to compare widely used multi-head and proposed multi-window approaches, which showed higher accuracy and interpretability of the multi-window approach.

The rest of the paper is organized as follows. Section 2 introduces preliminaries on graph attention and graph time series forecasting of PV generation. Section 3 details the proposed TSM-GAT architecture. The experimental results of our evaluation and the analyses are presented and discussed in Sections 4 and 5. Finally, we conclude in Section 6.

2. Multi-site PV power time series forecasting on graphs

The PV power forecasting task is a time series prediction task, predicting future PV power production, given past PV data. PV data is highly correlated in time since slowly varying local weather conditions

and cloud cover affect PV production across time steps. Further, cloud motion affects PV production of neighboring production sites sequentially, thus, making the data correlated in both spatial and temporal domains. Thus, part of the cloud dynamics influence can be inferred by modeling the past PV data as signals on a dynamic graph. This permits to capture of the dynamically changing spatio-temporal correlations in the PV data. Specifically, we denote the vector of PV power production over N PV stations at time step τ as $\mathbf{p}(\tau) \in \mathbb{R}^N$. The goal is to forecast $\mathbf{p}(\tau)$ for the next H discrete time steps ahead given M past observations:

$$\hat{\mathbf{p}}(\tau), \dots, \hat{\mathbf{p}}(\tau + H - 1) = f_\beta\left(\mathbf{p}(\tau - M), \dots, \mathbf{p}(\tau - 1)\right), \tag{1}$$

where for any τ, f_β is a chosen family of parametric estimators. A set of parameters β is learnt such that it minimizes the prediction error over the entire horizon by solving the following problem:

$$\arg\min_\beta \sum_{\tau \in \mathcal{T}} \sum_{\xi=\tau}^{\tau+H-1} \|\hat{\mathbf{p}}(\xi) - \mathbf{p}(\xi)\|_2^2, \tag{2}$$

where $\mathcal{T}$ represents the times of historical time steps which are used for fitting the model during training. Although we have chosen the mean square error as the metric, other metrics could be used as well.

In this work, we model the PV production time series as signals on a dynamic graph. The network of PV power stations is represented as a weighted undirected graph G^t with PV power production, where the graph is a tuple $G^t = (\nu, \varepsilon^t, \boldsymbol{\gamma}^t)$ and $t \in \{1, \dots, T\}$ represents the number of graphs in a dynamic graph. The graphs G^t and G^{t+1} are m time steps τ apart from each other. A set of vertices (nodes) $\nu = \{v_1, v_2, \dots, v_N\}$ in the graph corresponds to a set of PV stations and ε is a set of edges (links), corresponding to the correlations between the stations within the graph G^t. The most commonly used correlations are Pearson or distance correlation. The topology of the graph is described with its adjacency matrix $\boldsymbol{\gamma}^t \in R^{N \times N}$ that dynamically captures the connectivity of the nodes. Each adjacency entry represents an edge between v_i and v_j in the graph G^t, denoted with $e_{ij}^t \in \varepsilon^t$. If there is no edge between v_i and v_j in the certain graph G^t, then the matrix entry γ_{ij}^t is equal to zero. The dynamically changing correlations between PV stations are modeled with a dynamic adjacency matrix. Finally, we define a graph signal as a mapping $\mathbf{x} : \nu \to \mathbb{R}$, such that $x_v^t \in \mathbb{R}$ is the signal value at node v and graph G^t. For more details and an in-depth review of GSP we refer the reader to Ortega et al. [28].

In our case, the graph signal at time step τ is denoted by $\mathbf{x}(\tau)$ and it represents the vector of the power production of all PV stations $\mathbf{p}(\tau)$ concatenated with additional spatial information and geographical location. Furthermore, in order to provide the model with additional information regarding the sunrise, sunset and seasonality information, the clear sky irradiance values are additionally included $\mathbf{y} = (\mathbf{y}(\tau), \dots, \mathbf{y}(\tau + H - 1)) \in \mathbb{R}^{N \times H}$ for predicting H steps ahead, computed at a particular location on the map at any time, using the Ineichen and Perez clear sky model from PVlib [29]. This computation is completely deterministic and only relies on the geographical coordinates of the nodes (latitude, longitude and altitude). Thus we can reformulate our learning problem

$$\begin{aligned}\hat{\mathbf{p}}(\tau), \dots, \hat{\mathbf{p}}(\tau + H - 1) = f_\beta\big(&\mathbf{p}(\tau - M), \dots, \mathbf{p}(\tau - 1),\\ &\mathbf{y}(\tau), \dots, \mathbf{y}(\tau + H - 1), \mathbf{long}, \mathbf{lat}, \mathbf{U}\big),\end{aligned} \tag{3}$$

where $\mathbf{U}$ is a mask used to initialize the neighborhood of each node i. The initial neighborhood is calculated using k-nearest neighbor mechanism. The set of parameters β includes not only parameters for learning correlations in the dynamic graph, but also parameters for embedding the spatio-temporal features and finally parameters for predicting the future PV production.

3. Temporal-spatial multi-windows graph attention network

3.1. The overall architecture

In this paper, we propose a sequence to sequence model TSM-GAT. The model represents a solution for multi-site PV power forecasting from GSP perspective, shown in Fig. 1. A sequence of the past M PV measurements $\mathbf{p} = (\mathbf{p}(\tau - M), \dots, \mathbf{p}(\tau - 1)) \in \mathbb{R}^{N \times M}$ over N nodes is taken as input to the model, when predicting H steps ahead $\hat{\mathbf{p}} = \hat{\mathbf{p}}(\tau), \dots, \hat{\mathbf{p}}(\tau + H - 1)$. In order to capture both short- and longer-term dependencies, the attention mechanism from [30] is used since it has access to any part of the sequence.

TSM-GAT consists of temporal attention with the overlapping-window mechanism and spatial attention with the multi-window mechanism. The temporal attention operator is capturing temporal non-linear correlations using a modified attention mechanism. However, finding a correlation between all M past time steps leads to suboptimal solutions since it assigns a single attention value for all past time steps. Furthermore, weather conditions fluctuate and the wind might suddenly change directions, creating a challenge to model cloud dynamics. Therefore, the past M PV measurements are divided into T temporal windows. The past PV data in each temporal window is concatenated with vectors of geographical coordinates $\mathbf{long}, \mathbf{lat} \in \mathbb{R}^N$. The temporal attention is capturing temporal non-linear correlations for each temporal window. The embedded temporal features per each window represent entries of different graphs in the dynamic graph.

Thus, embedded features are passed to the spatial attention operator, which captures dynamically changing spatial correlations, yielding the dynamical adjacency. Finally, viewing the output of the temporal-spatial attention block as a vector encoding shading information, this vector is concatenated with an embedding of the clear-sky irradiance $\mathbf{y} = (\mathbf{y}(\tau), \dots, \mathbf{y}(\tau + H - 1)) \in \mathbb{R}^{N \times H}$ before the last multi-perceptron layer (MLP); see Fig. 1. The exact concatenation of PV power and clear sky irradiance signals is presented at the end of the section.

In order to calculate spatial attention, it is crucial to define the neighborhood of each node since it is not predefined. To this end, the adjacencies are initialized using the k-nearest neighbor algorithm. Intuitively, for the short-term forecasts (up to 2 h) the neighborhood taken into account should not be too large since the weather changes and cloud movements are affecting the nodes in the smaller localized area. As the forecasting window increases, nodes further away should be taken into account in order to capture cloud movement, since it takes more time for clouds to move between nodes that are spatially further away. Therefore, the multi-window mechanism is introduced, such that the different neighborhood is initialized for each window, where C is the total number of windows in the multi-window mechanism.

3.2. Graph attention

In this work, we use Graph Attention Networks (GAT) from [30] to infer the correlation between nodes. Let a signal value at node $i \in N$ be represented with a column vector $\mathbf{x_i} = [x_i^1, \dots, x_i^f] \in \mathbb{R}^f$, where f is the number of features per node. The attention mechanism is used to contextually weight the importance of node j features to the node i. A shared matrix $\mathbf{W} \in \mathbb{R}^{f' \times f}$ is used to embed input features f into a f'-level feature space. Then the normalized attention coefficients α_{ij} are computed from (4):

$$\alpha_{ij} = softmax_j\left(l\left(\mathbf{a} \cdot \left[\mathbf{W}\mathbf{x_i} \| \mathbf{W}\mathbf{x_j}\right]\right)\right) \tag{4}$$

where $\mathbf{a} \in \mathbb{R}^{2f' \times 1}$ is a row vector parameterizing the attention mechanism and $\cdot$ denotes dot product multiplication between vectors. A concatenation is represented with $\|$ and $l(\cdot)$ denotes the activation function LeakyReLU. Finally, the obtained normalized attention coefficients are used to compute the final output $\mathbf{h_i}$ for every node (5):

$$\mathbf{h_i} = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}\mathbf{x_j}\right) \tag{5}$$

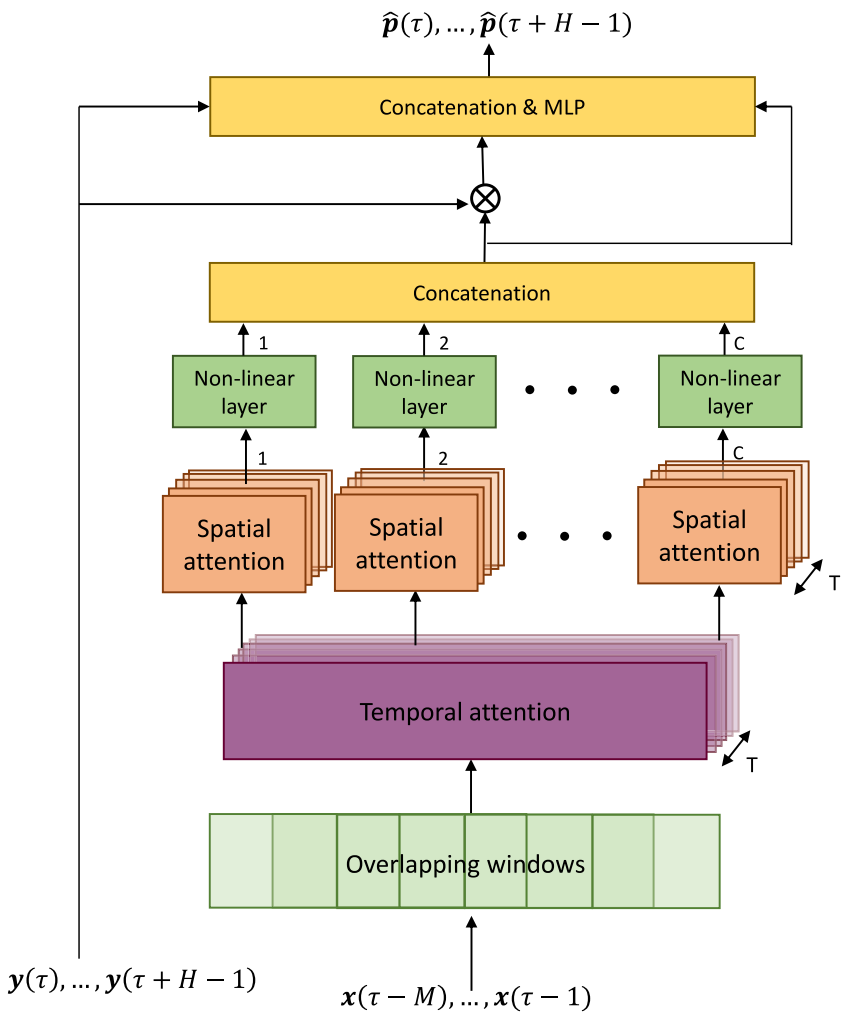


Fig. 1. TSM-GAT model.

where $\mathcal{N}_i$ represents the neighborhood of node i and $\sigma(\cdot)$ is the Leaky-ReLu nonlinearity.

3.3. Temporal attention

The temporal attention is capturing temporal non-linear correlations for each temporal window, by embedding the temporal features using modified attention. In the temporal attention the past data is divided in the T temporal windows. Then attention mechanism is applied in order to find the correlations between the temporal windows. Following a classical signal processing perspective [31], we have divided input feature space into the temporal windows with 50% overlap. The first half of each window is shared with the last half of the previous window, and the last half is shared with the first half of the subsequent one. Therefore, we divide the input M lags among T overlapping temporal windows of size $2m$. The window size is calculated based on the overlap m, where $m = \frac{M}{T+1} \in \mathbb{N}$. Since in each window longitude and latitude are concatenated as additional spatial information, the input signal is $\mathbf{x} = (\mathbf{p}(\theta), \mathbf{p}(\theta+1), \ldots, \mathbf{p}(\theta + 2\ m) \| \mathbf{long}, \mathbf{lat}) \in \mathbb{R}^{N \times T \times \mathrm{f_{in}}}$, where $f_{in} = 2\ m + 2$ and $\theta \in \{\tau - M, \tau - M + m, \tau - M + 2\ m, \tau - M + 3\ m, \ldots, \tau - 2m\}$.

In the work of Veličković et al. [30], in Eq. (4) the weight matrix $\mathbf{W}$ is shared across nodes. On the other hand, in the temporal attention of TSM-GAT, we created a matrix $\mathbf{W}^{\mathbf{t}}_{\mathbf{temp}} \in \mathbb{R}^{f' \times f_{in}}$ with different weight matrices for each time window $t \in \{1, \ldots, T\}$ to increase expressive power. Let $\mathbf{x}^{\mathbf{i}}_{\mathbf{t}} \in \mathbb{R}^{\mathrm{f_{in}} \times 1}$ denote the input signal for node i and temporal window t. We use the softmax function with $LeakyReLU$ activation function to get normalized temporal attention coefficients across all temporal windows:

$$\alpha^i_{kt} = softmax_t(l(\mathbf{a}^{\mathbf{t}}_{\mathbf{temp}} \cdot [\mathbf{W}^{\mathbf{t}}_{\mathbf{temp}} \mathbf{x}^{\mathbf{i}}_{\mathbf{k}} \| \mathbf{W}^{\mathbf{t}}_{\mathbf{temp}} \mathbf{x}^{\mathbf{i}}_{\mathbf{t}}])) \tag{6}$$

where the coefficient α^i_{kt} represents the importance of the temporal window k when making the prediction. A row vector $\mathbf{a}^{\mathbf{t}}_{\mathbf{temp}} \in \mathbb{R}^{2f' \times 1}$ is a weight vector of the attention mechanism calculated for each temporal window $t \in \{1, \ldots, T\}$ and shared across nodes. These normalized attention coefficients are used to compute the output of the temporal block $\mathbf{h} \in \mathbb{R}^{N \times T \times f'}$ where we define the column vector of features $\mathbf{h}^{\mathbf{i}}_{\mathbf{t}} \in \mathbb{R}^{f' \times 1}$, which is calculated for each node i and temporal window t:

$$\mathbf{h}^{\mathbf{i}}_{\mathbf{t}} = \sigma\left(\sum_{k=1}^{T} \alpha^i_{kt} \mathbf{Q}^{\mathbf{t}}_{\mathbf{temp}} \mathbf{x}^{\mathbf{i}}_{\mathbf{k}}\right) \tag{7}$$

where $\mathbf{Q}^{\mathbf{t}}_{\mathbf{temp}}$ is of the same size as $\mathbf{W}^{\mathbf{t}}_{\mathbf{temp}}$. The temporal attention matrix shows insight about which window influences the model when making the prediction for a certain horizon.

3.4. Spatial multi-windows attention

The TSM-GAT model captures temporal patterns across different time steps using the overlapping window mechanism with a temporal operator. However, in order to capture the correlation between different nodes at different time steps, a spatial graph attention operator is used on top of the temporal one. The input data to spatial attention are

embedded temporal features from different temporal windows. Those features represent entries of different graphs, creating the dynamic graph. Further, we refer to temporal windows as different parts of the input data, which embedded in the temporal attention are passed to the graphs in the dynamic graph.

Let the spatial attention tensor $\boldsymbol{\gamma}$ represent a sequence of dynamical spatial correlation matrices, where $\boldsymbol{\gamma} \in \mathbb{R}^{T\times N\times N}$. A single spatial attention matrix $\boldsymbol{\gamma}^t \in \mathbb{R}^{N\times N}$ is calculated for each temporal window $t \in \{1,\dots,T\}$. The correlation between nodes i and j at time window t is

$$\gamma_{ij}^t = softmax(l(\mathbf{a}_{\mathbf{spat}}^{\mathbf{i}} \cdot [\mathbf{W}_{\mathbf{spat}}^{\mathbf{t}}\mathbf{h}_{\mathbf{t}}^{\mathbf{i}} \| \mathbf{W}_{\mathbf{spat}}^{\mathbf{t}}\mathbf{h}_{\mathbf{t}}^{\mathbf{j}}])), \tag{8}$$

such that row vector $\mathbf{a}_{\mathbf{spat}}^{\mathbf{i}} \in \mathbb{R}^{2f''\times 1}$ have different values for each node $i \in \{1,\dots,N\}$. The weight matrix $\mathbf{W}_{\mathbf{spat}}^{\mathbf{t}} \in \mathbb{R}^{f''\times f'}$, in the spatial attention operator, is used to embed the input feature space f' to lower dimensional feature space f'' in each time window t, where $t \in \{1,\dots,T\}$. A spatial attention coefficient γ_{ij}^t indicates the importance of node j's features to the node i in the temporal window t. The spatial attention matrix represents the dynamical adjacency matrix since the attention coefficients are dynamically changing in each time window. In addition to spatial weight matrix $\mathbf{W}_{\mathbf{spat}}^{\mathbf{t}}$, the matrix $\mathbf{Q}_{\mathbf{spat}}^{\mathbf{t}} \in \mathbb{R}^{f''\times f'}$ where $t \in \{1,\dots,T\}$ is parameterizing the output of the temporal attention $\mathbf{h}_{\mathbf{t}}^{\mathbf{i}}$. The output of temporal-spatial attention block is a tensor $\mathbf{g} \in \mathbb{R}^{N\times T\times f''}$. We define a slice of the tensor $\mathbf{g}$ for node i and temporal window t as the column vector $\mathbf{g}_{\mathbf{t}}^{\mathbf{i}} \in \mathbb{R}^{f''\times 1}$ obtained from:

$$\mathbf{g}_{\mathbf{t}}^{\mathbf{i}} = \sum_{j\in\mathcal{N}_i} \gamma_{ij}^t \mathbf{Q}_{\mathbf{spat}}^{\mathbf{t}}\mathbf{h}_{\mathbf{t}}^{\mathbf{i}}, \tag{9}$$

where matrix $\mathbf{Q}_{\mathbf{spat}}^{\mathbf{t}}$, is of the same size as the matrix $\mathbf{W}_{\mathbf{spat}}^{\mathbf{t}}$. In $\mathbf{g}$, the important information is embedded in f'' distinctive features for each node and for each temporal window. Therefore, flattening the feature vectors $\mathbf{g}_{\mathbf{t}}^{\mathbf{i}}$ across temporal windows $t \in \{1,\dots,T\}$ yields an augmented matrix $\mathbf{G} \in \mathbb{R}^{N\times Tf''}$, where each row is a vector $[\mathbf{g}_{\mathbf{1}}^{\mathbf{i}}, \mathbf{g}_{\mathbf{2}}^{\mathbf{i}}, \dots, \mathbf{g}_{\mathbf{T}}^{\mathbf{i}}] \in \mathbb{R}^{Tf''}$ of embedded spatio-temporal features for the node i. This information is passed through a non-linear layer producing the sequence $\mathbf{s} \in \mathbb{R}^{N\times H}$, where H represents the length of the predicting horizon. Finally, viewing the output of the graph temporal-spatial attention block $\mathbf{s}$ as a matrix encoding the shading information, the clear-sky data is additionally introduced before the last MLP; see Fig. 1. Hence, $\mathbf{s}$ is concatenated with an embedding of the clear-sky data $\mathbf{y}$ and Hadamard product between $\mathbf{s}$ and $\mathbf{y}$.

The dynamical adjacency matrix $\boldsymbol{\gamma}^t$ of the graph is inferred in the spatial attention block using masked attention, in order to inject the graph structure. The adjacency values γ_{ij}^t in each graph are computed for all $j \in \mathcal{N}_i$, where $\mathcal{N}_i$ is the neighborhood of the node i on graph G^t. Thus, it is crucial to define the support of the neighborhood set $\mathcal{N}_i$ with a predefined mask. To this end, the mask is initialized using the Euclidean distance in the k-nearest neighbor (knn) algorithm. Thus, if the distance e_{ij}^t is larger than the one defined in knn algorithm, we will mask the adjacency entry γ_{ij}^t with 0. For different parts of the forecasting horizon H, the different neighborhoods are taken into account and different spatial attentions are calculated. Multi-window mechanism is introduced in the spatial attention, such that for short-, medium- and long-term dependencies we define $C = 3$ spatial attention windows. Thus, for each of the 3 windows, the k-neighborhood is considered to be $30\%, 50\%, 100\%$ of total number of nodes, respectively.

The spatial window c yields different spatio-temporal feature matrices $\mathbf{s}_{\mathbf{c}} \in \mathbb{R}^{N\times\frac{H}{C}}$ for $c \in \{1,\dots,C\}$. Let the matrix $\mathbf{G}_{\mathbf{c}} \in \mathbb{R}^{N\times Tf''}$ for window c be defined by stacking $t \in \{1,\dots,T\}$ feature vectors $\mathbf{g}_{\mathbf{t}}^{\mathbf{i,c}} \in \mathbb{R}^{f''}$ for node i. The vector of embedded features $\mathbf{g}_{\mathbf{t}}^{\mathbf{i,c}} \in \mathbb{R}^{f''}$ per window t and node i is given by:

$$\mathbf{g}_{\mathbf{t}}^{\mathbf{i,c}} = \sum_{j\in\mathcal{N}_i^c} \gamma_{ij}^{t,c} \mathbf{Q}_{\mathbf{spat}}^{\mathbf{t}}\mathbf{h}_{\mathbf{t}}^{\mathbf{i}}, \tag{10}$$

where dynamical adjacency entry $\gamma_{ij}^{t,c}$ is defined for each spatial window c is defined on different neighborhood $\mathcal{N}_i^c$. Corresponding different sets of weight matrices $\mathbf{W}_{\mathbf{spat}}^{\mathbf{i,c}} \in \mathbb{R}^{f''\times f'}, \mathbf{a}_{\mathbf{spat}}^{\mathbf{i,c}} \in \mathbb{R}^{1\times 2f''}$ are learnt for each window c. Hence, C spatial attention windows are calculated for different C parts of predicting horizon H which yields the feature matrix $\mathbf{s}_{\mathbf{c}}$:

$$\mathbf{s}_{\mathbf{c}} = \sigma\left(\mathbf{G}_{\mathbf{c}}\mathbf{B}_{\mathbf{c}} + \mathbf{b}_{\mathbf{c}}\right), \tag{11}$$

where $\mathbf{B}_{\mathbf{c}} \in \mathbb{R}^{Tf''\times\frac{H}{C}}, \mathbf{b}_{\mathbf{c}} \in \mathbb{R}^{\frac{H}{c}}$ represent weights and biases in the last linear layer of each spatial attention window $c \in \{1,\dots,C\}$. Finally, the feature matrix $\mathbf{S}$ is concatenation of all C feature matrices across all windows $S = [\mathbf{s_1}\| \dots \|\mathbf{s_c}]$. The feature matrix is only then concatenated with clear sky data. The predicted power for N nodes and H steps ahead is defined with

$$\hat{\mathbf{p}} = MLP\left([\mathbf{y}\|\mathbf{S}\|\mathbf{y}\odot\mathbf{S}]\right), \tag{12}$$

where the Hadamard product is denoted with $\odot$.

Thus, the TSM-GAT model finds dynamical adjacency which is different for each temporal window and for different predicting horizons. We can easily find the root cause and interpret on which nodes the model is focusing for short-, medium- and long-term forecasts within some given prediction horizon H. This approach makes the temporal-spatial multi-window graph attention model interpretable for the time series forecasting task.

3.5. Architecture configuration

We divide the input M lags among T overlapping temporal windows of size $2m$. The window size is calculated based on the overlap m, where $m = \frac{M}{T+1} \in \mathbb{N}$. One entire day of input data is chosen as the length M of the past observations, in order to find periodic dependencies and temporal shifts. The number T of the input overlapping sliding windows is 15 and the window length is 12 time steps which is equivalent to the past 3 h. Therefore, the size of the shift (and the overlap) between the two consecutive temporal windows is 1.5 h and these values have been chosen empirically. This choice is made with an assumption that the weather does not change drastically more than once within this horizon for each station. Thus, by embedding features in each window, we are embedding the change in the weather information each 1.5 h. In the spatial attention the output feature space is then divided into $C = 3$ different windows, thus, in each window the length of input clear sky data is $\frac{H}{3} \in \mathbb{N}$; see Fig. 1. The length of the window can be adapted in case where $\frac{H}{3} \notin \mathbb{N}$ such that an approximately similar number of time steps is in each window.

The k-neighborhood in the first window represents the closest 30% of the total number of nodes and it finds spatial correlations between the nodes focusing on the first two hours ahead prediction, i.e. it is assumed that only the closest neighbors contribute to first two hours prediction. The k-neighborhood in the second window is limited to 50% of the total number of nodes, focusing on the 2-4 h ahead prediction. The assumption for the second window is that the neighborhood should analyze nodes that are further away to consider longer term features. The neighborhoods of 30% and 50% of the total number of nodes are selected by hyper-tuning parameters to improve accuracy. Finally, the fully connected graph is used to find the correlations between nodes in the last time window that focuses on the 4 h to 6 h ahead prediction. The goal is to take into account both the closest and the furthest away nodes, when making predictions in the last window of the forecasting horizon. An MLP is used at the output of the second attention block to transform the temporal-spatial attention block outputs into the power production $\hat{\mathbf{p}} \in \mathbb{R}^{N\times H}$.

4. Experiments

We applied the TSM-GAT architecture on PV production data from real and synthetic datasets. Furthermore, we compared the performance of the TSM-GAT against state-of-the-art models for both single-

and multi-site forecasts on the real dataset. Single-site forecasts are used to compare results against models that use NWP data and with traditional model that uses only power data. On the other hand, multi-site forecasts are used in order to compare results with models that use multi-site PV power data, for both real and synthetic datasets. Then we analyzed what drives the performance of TSM-GAT in multi-site forecasts, on the real dataset, and conducted the study in order to compare the multi-window approach with the multi-head approach.

4.1. Datasets

The real dataset used in our study consists of records from 304 PV plants across Switzerland [9] for two years (2016–2017). The PV plants are spread inhomogeneously over the entire country, with a density reflecting the population density. The spatial distribution of this dataset is shown in Fig. A.8. The dataset has a 15-min resolution. The second dataset used is a synthetic dataset from NREL, with 405 PV stations distributed across California, USA, with 15-min resolution. This synthetic dataset consists of PV production data simulated from weather data for 1 year (2006) and is publicly available.[1] For both datasets, all available PV stations are used as both input and forecasting nodes.

4.2. Benchmark models

Five state-of-the-art methods were used as a benchmark in the multi-site forecasting evaluation. The first baseline is the spatio-temporal autoregressive model (STAR) [9] which is a linear autoregressive model. It uses the group Lasso regularization to promote sparse solutions, thus, creating the effect of selecting relevant nodes for the prediction of each site. The second and the third baselines are the recently proposed graph-based Graph convolutional long-short term memory neural network (GCLSTM) and Graph Convolutional Transformer (GC-Trafo) [17]. Both of these models have an encoder–decoder structure, however, the former is the recurrent-based graph convolutional model whereas the latter uses a transformer architecture coupled with graph convolution. The fourth baseline for multi-site forecasting is the non-graph-based space–time convolutional neural network (STCNN) [12]. It uses a greedy-adjoining algorithm to rearrange the stations based on their geographical proximity, before applying 2D convolution layers, as in image processing. The fifth baseline is TS-multi-head-GAT which represents modification of TSM-GAT where multi-head mechanism was used instead of multi-window.

For the single-site forecasts we used three state-of-the-art models and a simple smart persistence model as benchmarks. The first two models, which use NWP data, are Support Vector Regression (SVR) and the encoder–decoder long-short term memory network (EDLSTM). Both models use predictions of global irradiance and temperature as input weather data, which were obtained from Meteotest[2] with a temporal resolution of 1 h. Thus, in order to have 15 min resolution predictions, a simple sample-and-hold interpolation was applied to the NWP data. SVR is chosen as benchmark since [32] have shown that SVR outperforms several state-of-the-art models for intra-day forecasts. The second model, EDLSTM is a state-of-the-art encoder–decoder model, especially suited for time-series processing [33,34]. The EDLSTM uses past PV production data as well as NWP data to make a forecast. The encoder uses as inputs past observed weather and PV power data in order to estimate the state of the system. Then the decoder uses these estimations and NWP over the prediction horizon as input to the decoder. The third benchmark model is a Seasonal Auto-Regressive Integrated Moving Average with eXogenous factors (SARIMAX). The SARIMAX model uses past PV power and clear-sky irradiance as inputs.

[1] https://www.nrel.gov/grid/solar-power-data.html.
[2] https://meteotest.ch/en/.

4.3. Data preprocessing and training

The power data is normalized for both real and synthetic datasets by the maximum power production over the training year for TSM-GAT, TS-multi-head-GAT, GCLSTM, GCTrafo, SARIMAX, EDLSTM and STCNN. The STAR model, on the other hand, requires careful normalization in order to extract daily profiles from the historical data. The real dataset in Switzerland has two years of data, thus, for non-linear models, the first year (2016) is taken as training dataset, and the following year (2017) represents the evaluation dataset. On the other hand, the synthetic dataset in California has only one year of the data (2006). Therefore, the training dataset is from January until the end of July, and evaluation dataset is from August until December of 2006.

STAR and SARIMAX are linear autoregressive models, thus, in order to take into account seasonal weather changes, for the real dataset, the evaluation year (2017) is divided in small test batches. The models were trained over a period of data which is taken prior to each test batch. The STAR model coefficients were fitted over a period of two months and then used to predict the power of the next two weeks. SARIMAX coefficients were fitted over a period of three days and then used to predict power of the following day. The parameters for STAR and SARIMAX methods are re-fitted in a rolling window fashion for every two months and for every three days, respectively. The hyperparameters used in the TSM-GAT and multi-site baseline models can be found in Appendix. All models were trained on a workstation with 16 physical cores, 128 GB of RAM memory and an Nvidia RTX 2080 Ti GPU.

4.4. Evaluation and metrics

The model performance was assessed using several metrics. The peak normalized root-mean-square error (NRMSE) and the average normalized mean absolute error (NMAE) are defined as:

$$NRMSE(v,i) = \sqrt{\frac{1}{T}\sum_{\tau\in S}\left(\frac{\hat{p}_v(\tau+i)-p_v(\tau+i)}{p_v^{max}}\right)^2}, \tag{13}$$

$$NMAE(v,i) = \frac{\sum_{\tau\in S}|\hat{p}_v(\tau+i)-p_v(\tau+i)|}{\sum_{\tau\in S}p_v(\tau+i)} \tag{14}$$

where $p_v(\tau)$ and $\hat{p}_v(\tau)$ denote the ground truth power and predicted power, respectively, of site v at time τ and horizon i. p_v^{max} is the maximum power of site v over the evaluation period S, i.e., the 2017 year, and T is the number of time steps in the evaluation interval S. Night times are excluded from error computations.

The second type of metric represents the difference in the shape between the predicted power and the ground truth. Dynamic time wrapping (DTW) is used as a measurement of the distance between shapes in time series. DTW distance $\delta_v(\tau+i,\tau+j)$ for node v and between time series $\hat{p}_v(\tau),\dots,\hat{p}_v(\tau+i)$ and $p_v(\tau),\dots,p_v(\tau+j)$ is defined as the minimum of accumulated distances $dist(p_v(\tau+i),\hat{p}_v(\tau+j))$:

$$\begin{aligned}\delta_v(\tau+i,\tau+j) &= dist(\hat{p}_v(\tau+i),p_v(\tau+j))+\\ \min\{\ &\delta_v(\tau+i-1,\tau+j),\delta_v(\tau+i,\tau+j-1),\\ &\delta_v(\tau+i-1,\tau+j-1)\ \}\end{aligned} \tag{15}$$

where Euclidean distance was used as the distance measurement. DTW is an indication of how good a model is at predicting shapes, especially clouds since they largely affect the shape of production. For calculations we used the approximation of DTW with linear time complexity presented in [35].

We classified days based on production variability into sunny, cloudy and variable in order to analyze the impact of weather conditions on the error rate and shape difference of time series, following definition from [36]. The main difference is that instead of irradiance in our case the input data is PV power data. Two metrics were used for

this classification: daily aggregate ramp rate (DARR) and daily index K_v^d, defined in [37]. The DARR is a simple metric given by

$$DARR_d = \sum_{\tau \in I_d} \frac{|p_v(\tau) - p_v(\tau+1)|}{p_v^{max}}, \tag{16}$$

where I_d represents the set of the daily values (night values are not taken into account) during the day d. Days when $DARR_d > 3$ are classified as variable days, caused by having sun at some parts of the day and then clouds for the rest of the day, with sudden peaks and drops, which is essentially causing the high variability. However, the DARR metric has one limitation when it comes to classifying sunny and overcast days since overcast days show small rates, like the sunny days with $DARR_d < 3$. Therefore, to account for overcast days the additional metric with the daily index is introduced. For each time step the daily index K_v^d for each node and each day d is calculated

$$K_v^d = \frac{1}{|I_d|} \sum_{\tau \in I_d} \frac{p_v(\tau)}{y_v(\tau) p_v^{max}} \tag{17}$$

where $y_v(\tau)$ is the normalized clear-sky irradiance at site v. As a threshold value for classification we chose $K_v^d = 0.3$, where values above the threshold are indicate a sunny day, whereas lower values imply that the day in question is cloudy.

5. Results

5.1. Prediction accuracy

In Fig. 2 the NRMSE and NMAE evolution of the TSM-GAT is compared to the STAR, GCLSTM, GCTrafo and STCNN models over a predicting horizon of 6 h ahead, with a 15-min resolution for the real dataset in Switzerland. The shaded regions represent the inter-quartile (25%–75%) error range and solid lines represent the median of error over all nodes. Although the GCLSTM method is on par with the TSM-GAT for the first hour ahead prediction, the TSM-GAT outperforms the GCLSTM from four to six hours ahead predictions. The region of interest, between four and six hours ahead of prediction, is highlighted and shown magnified on the right. Furthermore, the lower error slope and high accuracy in the 5th and 6th hour ahead prediction of the TSM-GAT could make it a promising multi-site model for longer prediction horizons. NRMSE is more sensitive to outliers since it considers squared errors. Therefore, NMAE is introduced in addition and the error evolution of NMAE is visually shown in Fig. 3(a). The GCLSTM still has the lowest NMAE for 1 to 4 h ahead predictions. However, the TSM-GAT has lower errors than any other state-of-the-art model for 4 to 6 h ahead forecast on both metrics. Since the error at the end of the forecasting horizon becomes higher, the advantage of TSM-GAT in capturing long-term correlations becomes noticeable.

In Fig. 3 the NRMSE and NMAE evolution of the TSM-GAT is compared to the non-linear methods, GCLSTM, GCTrafo and STCNN, over a prediction horizon of 6 h, with a 15-min resolution for synthetic dataset in California. The authors of STCNN, Jeong and Kim [12], report lower error in their paper, since they use only a subset of 238 nodes and hourly temporal resolution. Thus, they average the data which leads to smoothing signals and lower error rate. Similarly to the performance on the real dataset, the TSM-GAT outperforms state-of-the-art non-linear methods for 4 to 6 h ahead prediction on the synthetic dataset. Thus, the region of interest, between four and six hours ahead of prediction, is highlighted and shown magnified on the right. TSM-GAT has lower error slope, while staying on par with GCLSTM in the first four hours of the predicting horizon. The performance of all models on the synthetic dataset shows lower error compared to the real dataset. This is the result of smoothing in the synthetically created PV power dataset, generated from irradiance data using the Sub-Hour Irradiance Algorithm [38].

We also compare forecasting results for one site in the central part of Switzerland, Bätterkinden which is close to Bern. Fig. 4 shows the NRMSE evolution for the proposed method and benchmark single-site models. The error of the persistence model is up to 25% in the last hour of prediction. However, the NRMSE of persistence model is clipped at 20% in Fig. 4 in order to be able to see the error of other models more clearly. The error rate in single-site comparison shows that TSM-GAT outperforms all single-site models. The exception is in the sixth hour ahead, where for the last four steps of the horizon (21st to 24th step ahead), EDLSTM yields lower error than the proposed method. Although EDLSTM has higher error from 0 to 5 h ahead, compared to the proposed TSM-GAT, it clearly benefits from NWP information at the end of the horizon since NWP data improves predictions from 6 h ahead to one day ahead.

Table 1
Shape distance of predicted and observed time series using DTW.

Prediction ahead:	Weather conditions	TSM-GAT	GCLSTM	TS-multi-head-GAT
1 h	Overall	**1.42**	1.69	1.69
	Sunny	**1.15**	1.36	1.40
	Cloudy	**0.65**	0.92	0.83
	Variable	**2.87**	3.24	3.26
3 h	Overall	**1.86**	2.08	2.08
	Sunny	**1.52**	1.64	1.76
	Cloudy	**1.18**	1.46	1.24
	Variable	**3.37**	3.61	3.72
6 h	Overall	**2.47**	2.72	2.75
	Sunny	**2.21**	2.24	2.50
	Cloudy	**1.93**	2.56	2.21
	Variable	**3.69**	3.81	3.98

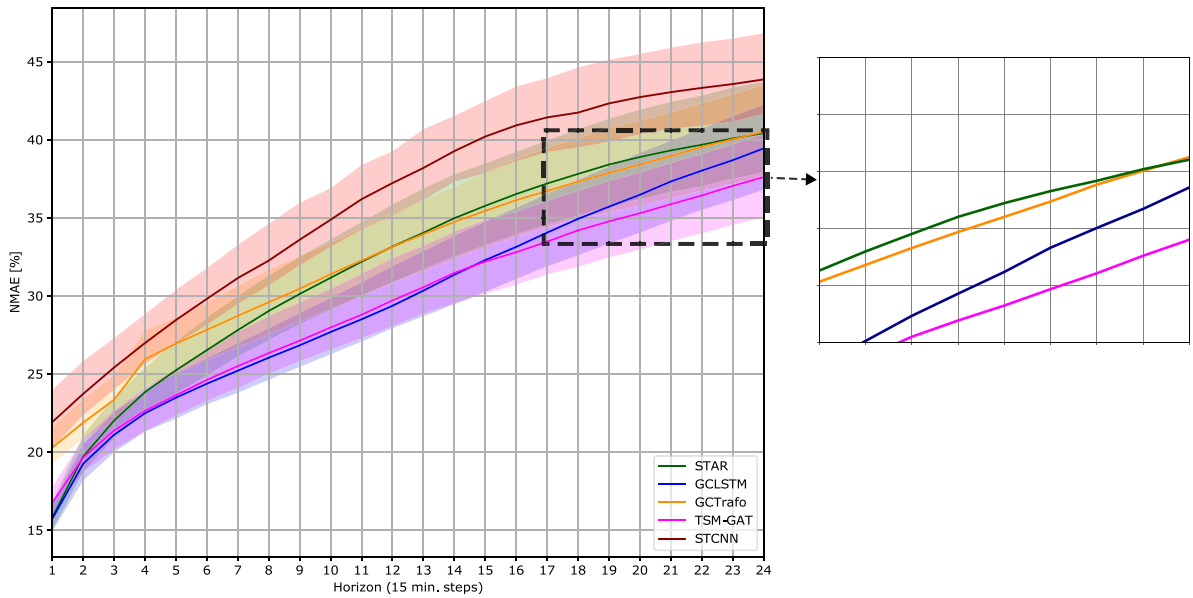
5.2. Comparative analysis

In order to better understand what drives each model, discover pitfalls or particular advantages we use the DTW metric in addition to NRMSE and NMAE. We make comparison only with the real dataset, in order to analyze better the impact of weather conditions on PV power forecast, since on the synthetic dataset the signals are smoothened out and the sharp weather transitions are less pronounced. Furthermore, the comparison is focused only on the GCLSTM and TSM-GAT since they have the lowest errors. Table 1 illustrates the similarity in shape between time series, obtained based on DTW metric for sunny, cloudy and variable days. Despite the lower NMAE error for the GCLSTM in the first two hours of the prediction, TSM-GAT has lower DTW values during cloudy and variable days in Table 1 as well as lower NRMSE on 1 to 6 h ahead forecasting horizon. This suggests the ability of TSM-GAT to accurately predict abrupt changes in the cloud movement or day to day weather. Hence, it yields a daily time series shape closer to the ground truth, which is important for energy management application.

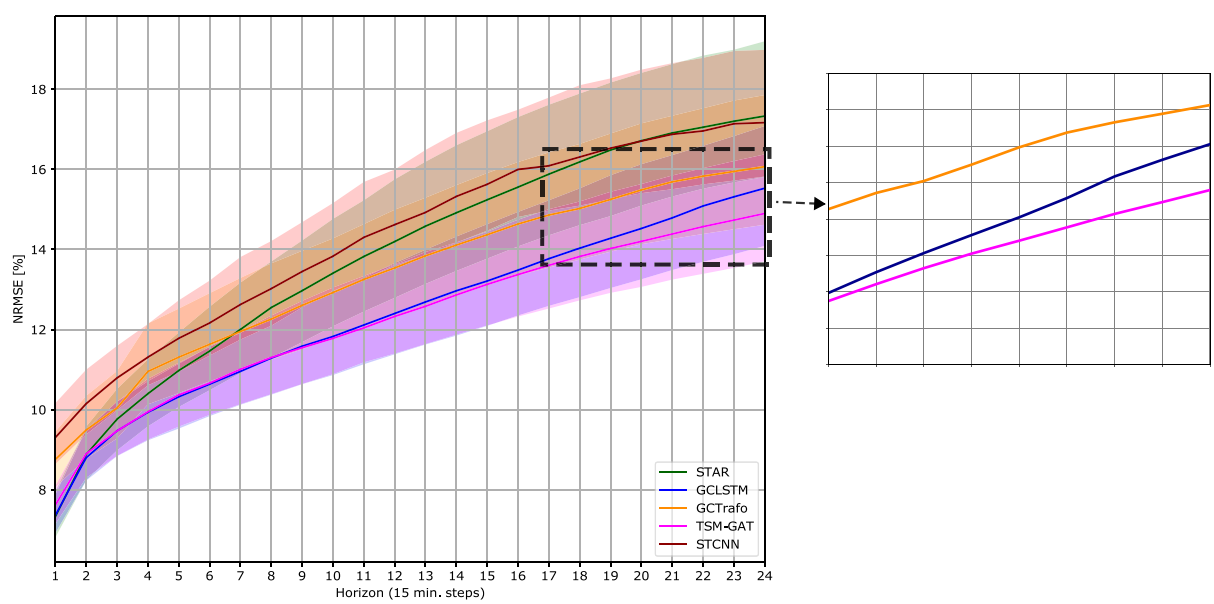
We analyze the correlation between the NRMSE and the number of variable, sunny and cloudy days. Nodes with the higher number of variable days have higher error for 6 h ahead prediction, whereas for the sunny and cloudy days, there is no clear dependence. The analysis of the error with respect to the distance to the centroid indicate that methods performs similarly for the central nodes and on the edges on the graph. Furthermore, the analysis of the error with respect to average distance to the closest neighbors shows that being in a cluster or being isolated, does not affect the performance of the model. The results are shown and discussed in more detail in Appendix B, along with the limitations of the model.

5.3. Effect of multi-window approach

A study has been conducted on the same dataset to verify the hypothesis regarding the higher accuracy and interpretability of the proposed multi-window attention against multi-head attention. We modified TSM-GAT by replacing the multiple windows with multiple



(a) Forecast NMAE for the real dataset.

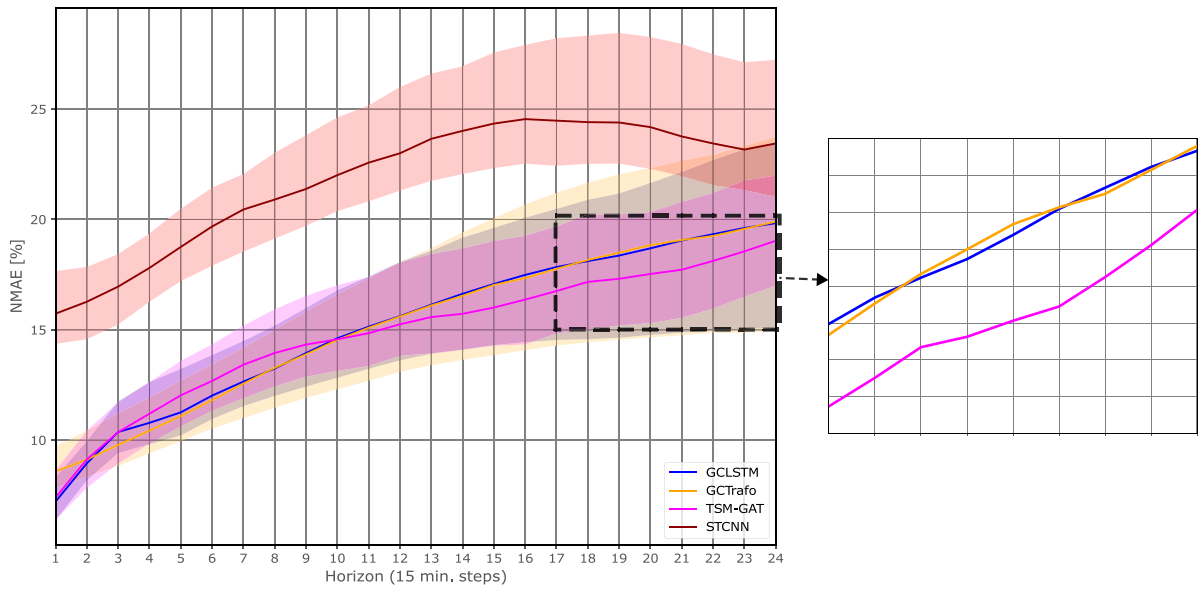


(b) Forecast NRMSE for the real dataset.

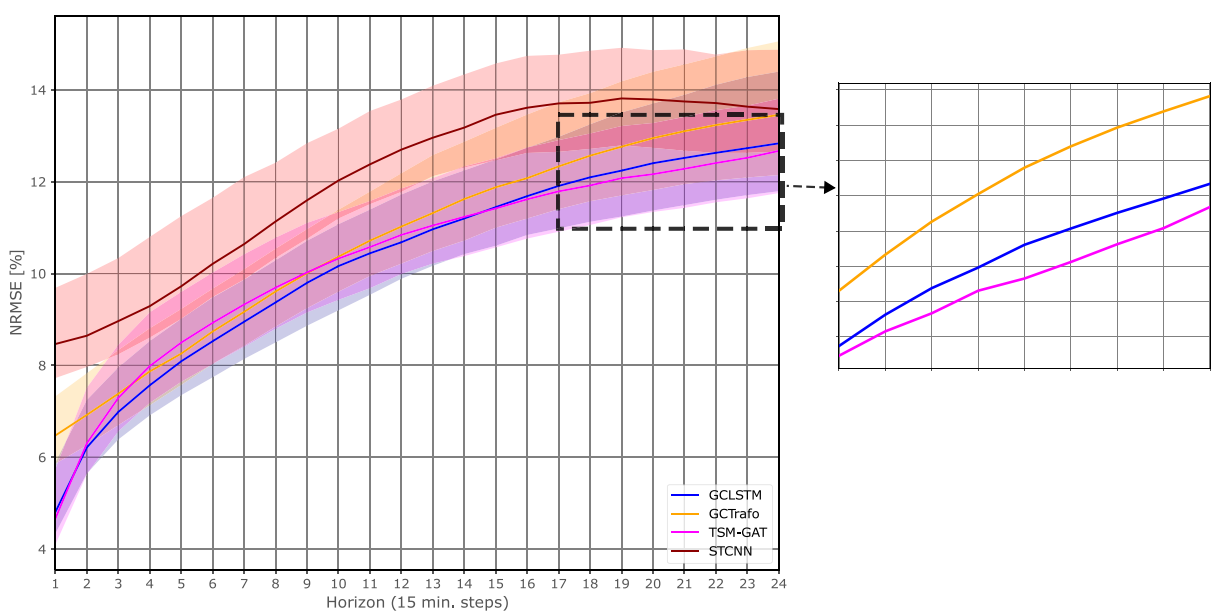
Fig. 2. Error comparison of TSM-GAT and state-of-the-art models for six-hour ahead prediction for the real dataset in Switzerland with magnified part between 4 and 6 h ahead prediction. Solid line shows the median value among all nodes, shaded bands show the interquantile range among all nodes.

heads, creating the temporal-spatial multi-head graph attention mechanism (TS-multi-head-GAT). Here the TSM-GAT and TS-multi-head-GAT are evaluated. Although the GCTrafo model also uses a multi-head approach, its model architecture is completely different compared to the TSM-GAT. Moreover, it is reporting higher error compared to both temporal-spatial GAT architectures (multi-window and multi-head), thus, it has not been used in this study. The main difference between TSM-GAT and multi-head approach is that the TSM-GAT has a multi-window approach where each window focuses explicitly on one part of the predicting sequence, whereas the multi-head approach focuses on the entire predicting sequence at once. The number of attention heads and attention windows in this part of the study is the same: $C = 3$.

The first advantage of the TSM-GAT model is learning lower number of parameters in the spatial attention, in the last linear layer before concatenation with clear-sky data. The size of the weights and biases in each attention window is $\mathbb{R}^{Tf''\times\frac{H}{C}}$, $\mathbb{R}^{\frac{H}{C}}$, respectively, since it is embedding feature sequence Tf'' into the length of the future window sequence $\frac{H}{C} \in \mathbb{N}$. On the other hand, in the multi-head approach, with C heads, the sizes of weights and biases in this linear layer are $\mathbb{R}^{Tf''\times H}$, $\mathbb{R}^{H}$. Moreover, multi-window approach requires an additional layer, to reduce the number of concatenated features from CH to H



(a) Forecast NMAE for the synthetic dataset.



(b) Forecast NRMSE for the synthetic dataset.

Fig. 3. Error comparison of TSM-GAT and state-of-the-art models for six-hour ahead prediction for the synthetic dataset in California with magnified part between 4 and 6 h ahead prediction. Solid line shows the median value among all nodes, shaded bands show the interquantile range among all nodes.

in order to make it possible to integrate the clear-sky data. On the other hand, TSM-GAT has the advantage of focusing on the different parts of the input sequence due to the fact that windows do not have overlap in specializations. Trainable parameters and defined variables are dependent on the number of sliding windows T, number of features f_{in}, f', f'', number of multi-head windows C and total number of nodes N. Although we have a restricted number of neighbors on which the model focuses when making the prediction, a full matrix–vector multiplication is calculated to find the correlation between the nodes in the dynamical adjacency matrix γ^t. Only then the entries which are not in the predefined neighborhood are masked with zeros. Since sparse matrix structures are not used, the number of the neighbors does not affect the computational complexity. Therefore, computational complexity and memory requirements scale with $\mathcal{O}(N^2T)$.

To better understand the model we focus on the forecasting results from a specific moment in time and place. The analysis was made on 30th of August, which was the cloudy day, at 7 a.m. for the next 6 h ahead prediction, which is the third day in Fig. 5. We choose this day since it is cloudy one after the sunny days and this time since it is difficult to make forecast in the morning when the most recent past data is limited to only one hour of data and then followed by night values. We analyzed the spatial and temporal adjacencies in detail for this example in Appendix C.

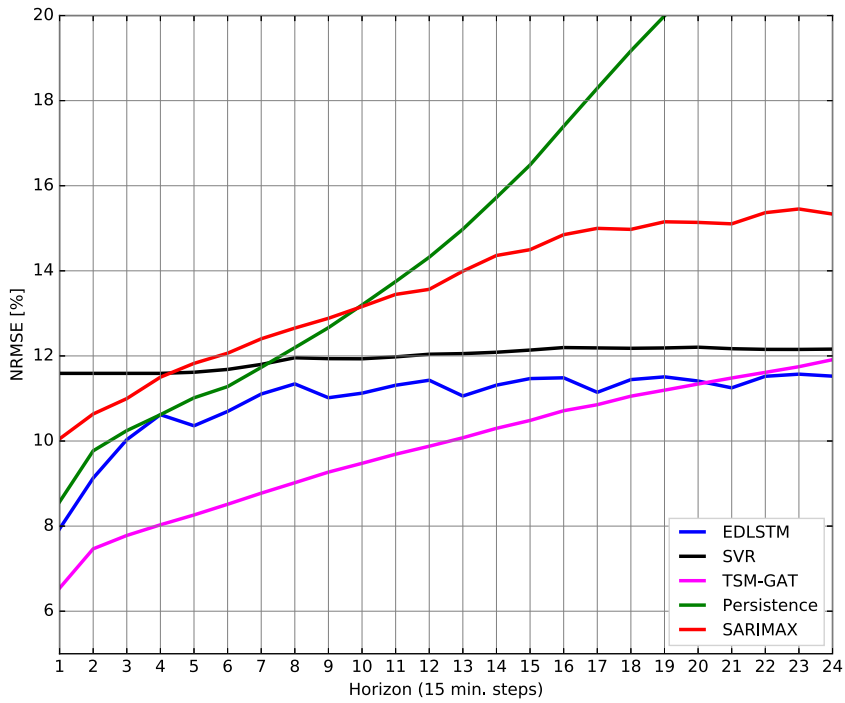


Fig. 4. Single-site error comparison between the TSM-GAT and state-of-the-art single-site models.

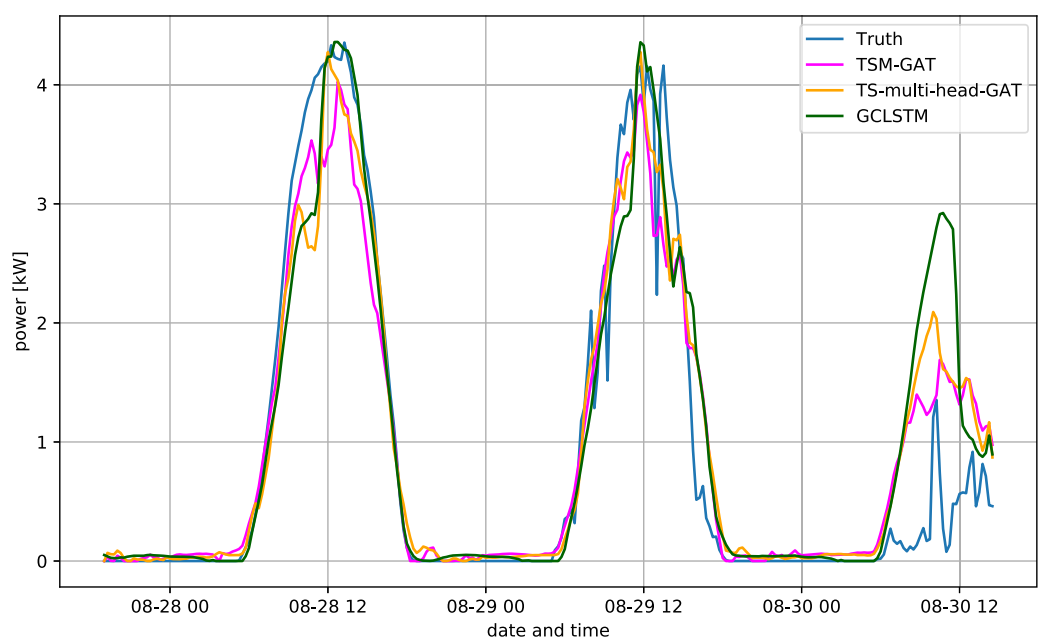
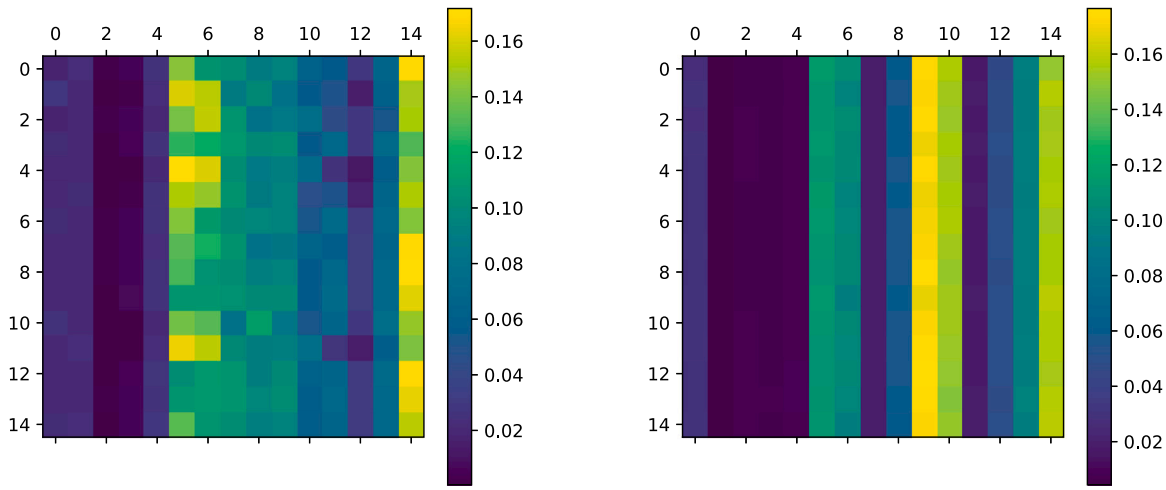


Fig. 5. Ground truth and prediction for 6 h ahead.

The temporal adjacency is shown in Fig. 6. In this example the temporal attention of multi-window model focuses on the last temporal window before the prediction and on 2 windows just before the sunset of the previous day, which is intuitive. On the other hand, the multi-head model focuses on the last temporal window, but it also focuses on the temporal windows during night, which should not be relevant part of the input sequence.

The analysis of the full dynamical adjacency for only one snapshot in time with 15×304^2 entries would be difficult. Therefore, we will focus on the 15th window of adjacency, since for both models temporal attention is high in this window. Nodes above an arbitrary threshold are considered important. In Table 2 the number of important nodes is shown, as well as the percentage of these important nodes from each window/head, which also represent an important node in the other head(s)/window(s). The percentage of the important nodes per head, which is shared with other head(s), is much higher than the percentage of nodes per window, shared with the other window(s). Thus, it is more difficult to interpret the results in the multi-head attention due to the high overlap. On the opposite, in the multi-window approach, each window has a lower percentage of the overlap. TSM-GAT benefits from the clear specialization since with the increase of forecasting horizon the average distance between the observed node and those to which the mechanism is focusing on increases as well, see Fig. 7.



(a) TSM-GAT. (b) TS-multi-head-GAT.

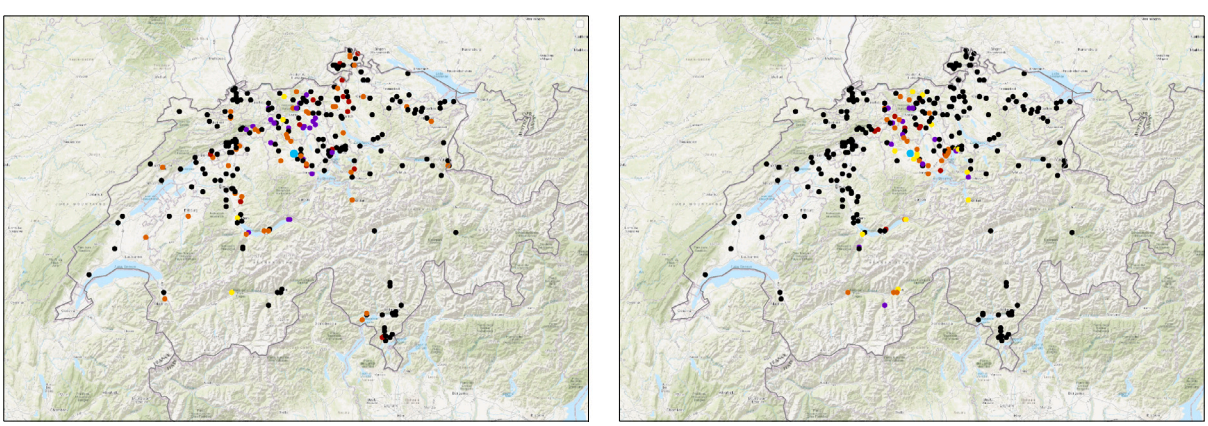
Fig. 6. Temporal attention between 15 overlapping windows. Darker colors signal lower attention.

Table 2
Analysis of the nodes with the highest spatial attention coefficients.

Important nodes	Head/Window 1		Head/Window 2		Head/Window 3	
	Nodes	Shared [%]	Nodes	Shared [%]	Nodes	Shared [%]
TS-multi-head-GAT	37	39%	54	54%	23	30%
TSM-GAT(multi-window)	30	16%	19	10%	52	9%

Table 3
Forecasting performance of TSM-GAT and baseline models.

	15 min		1 h		2 h		3 h		4 h		5 h		6 h	
	NRMSE	NMAE	NRMSE	NMAE	NRMSE	NMAE	NRMSE	NMAE	NRMSE	NMAE	NRMSE	NMAE	NRMSE	NMAE
GCLSTM	**7.36**	**15.70**	**9.93**	22.48	**11.29**	**26.04**	12.40	**29.36**	13.48	33.13	14.52	36.47	15.53	39.44
TSM-GAT	7.61	16.74	**9.94**	22.63	**11.30**	26.33	**12.32**	29.70	**13.37**	**32.79**	**14.20**	**35.29**	**14.89**	**37.60**
TS-multi-head-GAT	7.57	16.80	9.99	**22.44**	11.40	26.20	12.47	29.92	13.72	33.63	14.80	36.32	15.72	38.95



(a) TSM-GAT spatial attention. (b) TS-multi-head-GAT spatial attention.

Fig. 7. Map of the spatial attention in the last overlapping window. The prediction is made for the node in turquoise color. Attention coefficients below the threshold are black. The purple, red and orange nodes have coefficient values above the threshold in the first, second and third attention head/window, respectively. Yellow nodes have values above threshold on which at least two out of three heads/window focus (shared between heads).

Finally, we compare errors of TSM-GAT, TS-multi-window-GAT and GCLSTM in Table 3. According to James Murdoch et al. [39] interpretability is defined as predictive accuracy, descriptive accuracy, and relevancy, where relevancy is judged by a human audience. We measured the predictive accuracy by using NMAE, NRMSE in Table 3. Overall the highest accuracy is for the TSM-GAT model. What is more, the descriptive accuracy is higher with the TSM-GAT since we can understand on which nodes the model is focusing for different parts of the forecasting horizon. However, this is not the case with the multi-head approach where the model is focusing on the same nodes when making 1 h ahead and 6 h ahead prediction. On the other hand, with recurrent structures in the GCLSTM it is extremely difficult to find where is mechanism focusing, since it required tracking the activations and checking their updates. Finally, relevancy is the last interpretation measure where DTW metric in Table 1 indicates lower shape difference to the ground truth of the TSM-GAT, which is making this model

more relevant for energy management application. Thus, this suggests that the TSM-GAT model with the multi-window approach is more interpretable than the multi-head mechanism or state-of-the-art model GCLSTM.

6. Conclusions

A novel method, TSM-GAT, for capturing dynamically changing spatio-temporal correlations in deterministic PV power forecasting has been introduced. It was evaluated on real and synthetic PV production datasets and the performance was compared against state of the art, for both multi-site and single-site models. The TSM-GAT outperformed state-of-the-art methods from 4 to 6 h ahead prediction on the real and synthetic datasets. Results suggest that the TSM-GAT model is better than the state of the art at capturing shadowing, including cloud motion, and weather changes since it yields signal shapes closer to the ground truth on the entire horizon. Thus, it is addressing the limitation of cloud prediction in the spatio-temporal task, by directly capturing dynamically changing adjacency matrices for different parts of predicting horizon. A study was conducted to analyze the difference between the widely used multi-head attention and the proposed multi-window attention. It indicates that the proposed TSM-GAT model is more interpretable, and therefore more suitable for the prediction of multi-site time series driven by physical phenomena, such as PV and wind forecasting. This model could be used for other time series forecasting tasks where physical phenomena are modeled or when it is important to understand what influences the forecast. An in-depth analysis was performed in order to better understand what drives the performance on PV dataset. It is shown that the distance between PV power plants does not affect accuracy of the forecast, which is important for countries where PV stations are not homogeneously and densely distributed. Among the investigated state-of-the-art methods, the TSM-GAT model has the lowest distance between predicted and ground-truth signal shapes, as well as the lowest error for 4 to 6 h ahead forecasts, which is important for energy trading and energy management. On top of that, the lower dependence of its error on the time horizon makes TSM-GAT very promising for longer-term (day-ahead) predictions.

The architecture is scalable only for hundreds of nodes and the scalability limitation will be addressed in future work. Although we have shown here that the method does not have limitation in terms of distance between power plants, this finding should be confirmed in further research on a denser dataset containing a higher number of homogeneously spread nodes. Furthermore, robustness against missing data, which represent a common problem in time series forecasting, is yet to be investigated.

CRediT authorship contribution statement

Jelena Simeunović: Conceptualization Methodology, Software, Validation, Formal analysis, Writing – original draft, Investigation, Resources, Data curation, Writing – review & editing, Visualization. **Baptiste Schubnel:** Conceptualization, Software, Resources, Investigation, Data curation, Validation, Writing – review & editing, Resources. **Pierre-Jean Alet:** Funding acquisition, Validation, Writing – review & editing, Resources, Validation, Project administration. **Rafael E. Carrillo:** Conceptualization, Methodology, Writing – review & editing, Validation, Resources, Supervision, Project administration. **Pascal Frossard:** Conceptualization, Methodology, Writing – review & editing, Validation, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors use real and synthetic datasets. The synthetic dataset is publicly available, and we have shared the link to this data. However, the real dataset is confidential.

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Appendix A. Hyperparameters

The hyperparameters used to train baseline models STAR, STCNN, GCLSTM and GCTrafo could be found in works of Simeunović et al. [17] and Carrillo et al. [9]. STCNN and STAR use only PV power production data as input to the model, whereas GCLSTM, GCTrafo, TSM-GAT and SARIMAX use longitude and latitude as input, to create graph and compute clear-sky irradiance, in addition to PV power production. SVR uses NWP and clear-sky data as input, whereas EDLSTM uses PV power production in addition to NWP data. SARIMAX uses a seasonality order which is equal to the number of days to which data is fitted. The number of past lags that is fed to the model is 12, which corresponds 3 h of data. The difference operator order is 1 and moving average operator is of order 3. EDLSTM was trained with a dropout rate of 0.05 as regularization for 30 000 iterations. The number of past lags is 16, which is 4 h of data, with batch size 128 and learning rate $1e{-}4$. As far as proposed TSM-GAT is concerned, number of hidden features in temporal attention is reduced from $f_{in} = 14$ to $f' = 8$, and then in spatial attention to $f'' = 4$ features per temporal window and per node. This hyperparameter was decided arbitrary, as well as the number of the closest neighbors, which is changing between overlapping windows as already described. The model was trained with batch size 32 and for 600 000 iterations for the real dataset and 680 000 iterations for the synthetic dataset. Layer normalization is performed after spatial attention. The model was trained with stochastic gradient descent and Adam optimizer and learning rate 1e−4. The size of hidden layer in MLP used for final prediction is 64. Model TS-multi-head-GAT has the same sizes of the temporal weights as TSM-GAT. The number of closest neighbors for TS-multi-head-GAT is fixed to 90 for all attention heads. The number of iterations is the same as in TSM-GAT. Three heads were used and number of the weights in the linear layer is 3 times higher than in TSM-GAT, as already discussed.

Appendix B. Analysis and limitations of the model

Both GCLSTM and TSM-GAT display a similar behavior in terms of correlation between error and the number of variable days per node. Fig. A.8 shows the number of variable days in a year per node, where darker colors indicate the lower number of variable days per node. Several nodes are in a dense cluster, and yet have more variable days than the rest of the nodes in their clusters. This indicates that they could have different micro-climate, shadowing effects or more foggy days than the rest of the nodes in their clusters. At these nodes both GCLSTM and TSM-GAT showed higher error for six hours ahead prediction; see Fig. B.1. Thus, the first limitation of the model is higher error in case of the nodes with more variable days than the rest of the nodes in the cluster. Relying on the neighborhood information increases the error when the predicting nodes have much higher number of variable days than the close neighbors in their clusters. This represents an indication that the models are not able to capture the specific microclimate at these distinctive nodes. As expected, from Fig. B.2 we can see that the higher the number of variable days per node, the higher the error per node. However, there is no clear dependence between the error and the number of sunny or cloudy days.

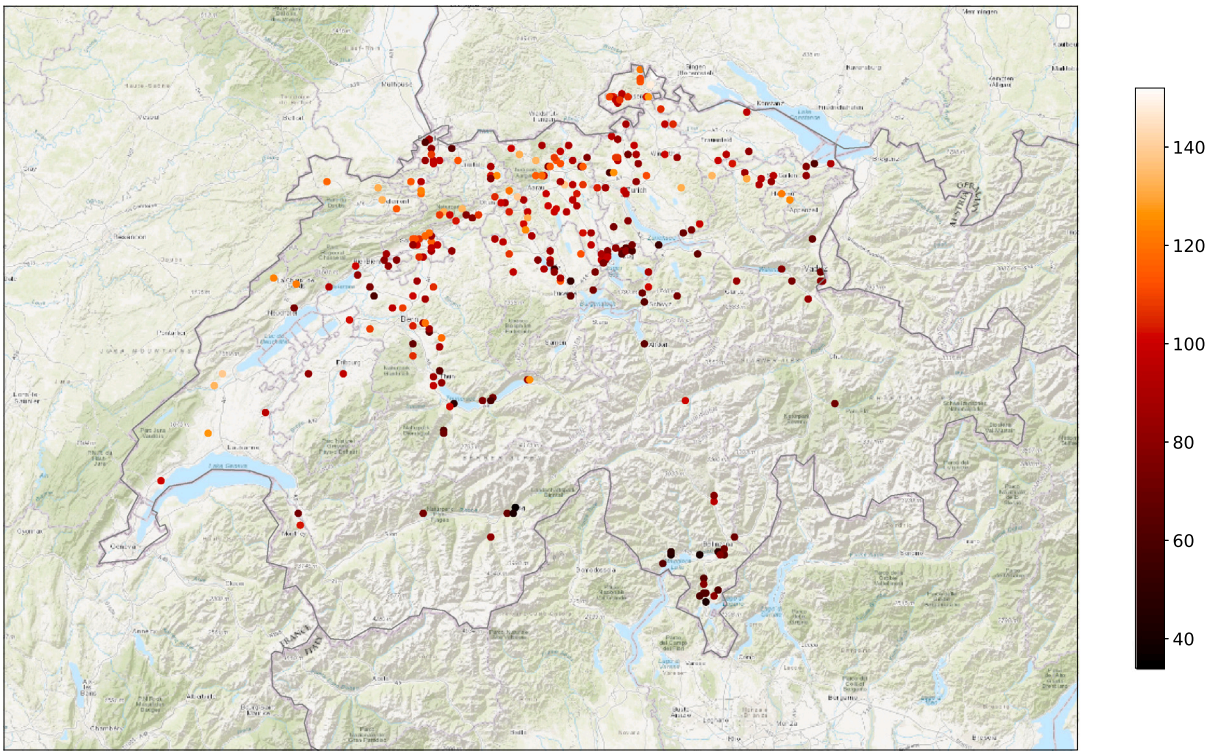

Fig. A.8. Number of variable days per node. The darker colors are signalizing the lower number of variable days in year 2017 per node.

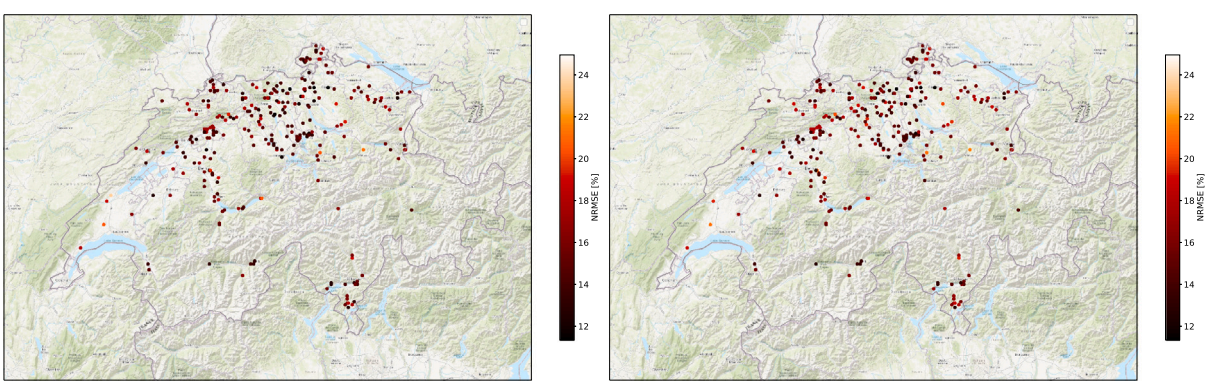

Fig. B.1. NRMSE for 6 h ahead prediction per node (in [%]) in the year 2017 for TSM-GAT (left) and GCLSTM (right). Darker colors are indicating the lower NRMSE per node.

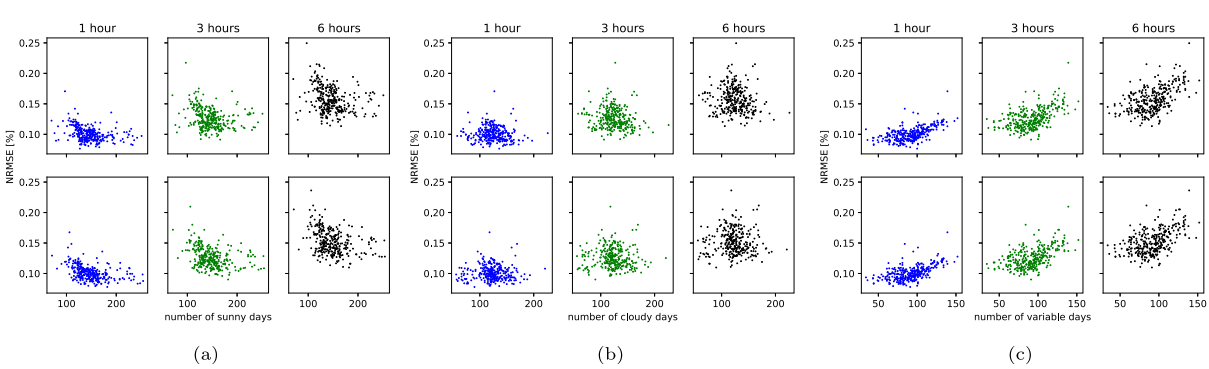

Fig. B.2. NRMSE of TSM-GAT model (bottom) and GCLSTM model (top) for 1,3,6 h ahead prediction per node for 2017 year divided into 3 different type of days. (a) Sunny days. (b) Cloudy days. (c) Variable days.

Another limitation of the model is high computational complexity compared to GCLSTM, since the computational complexity of the TSM-GAT model scales with $\mathcal{O}(N^2T)$ where N is the total number of nodes, T is the number of temporal windows. Oppositely, GCLSTM computational complexity scales with $\mathcal{O}(Nkl)$, where k is the number of closest neighbors in GCLSTM, and l is the number of time steps taken into

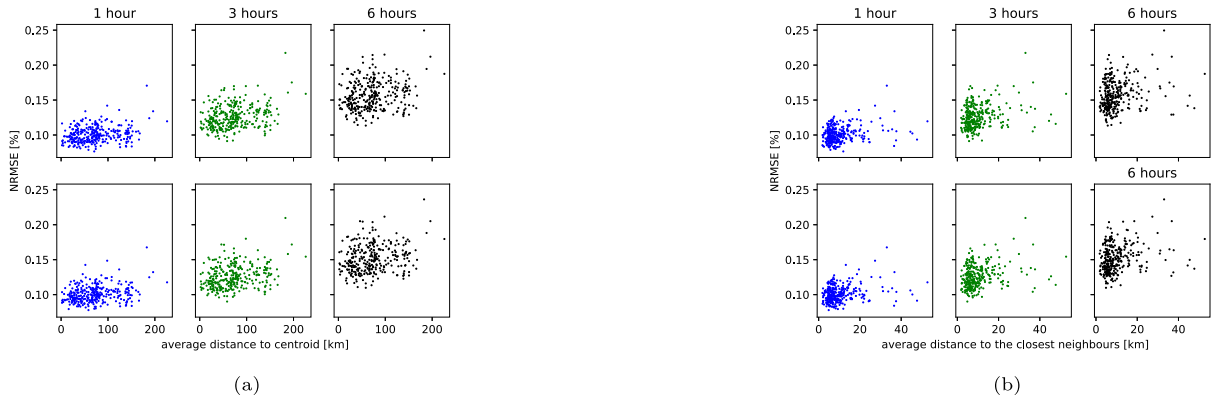

Fig. B.3. NRMSE per node for 1,3 and 6 h ahead prediction with respect to different distances for TSM-GAT (bottom) and GCLSTM (top). (a) NRMSE with respect to distance to the centroid. (b) NRMSE with respect to distance to 5 closest neighbors.

account for recurrence, such that $NT \gg kl$. Furthermore, in TSM-GAT model the number of learned weights is T times higher, increasing the memory requirements.

The analysis of the error with respect to distance from the centroid, where the centroid is calculated as the average of nodes' coordinates, does not indicate a higher error for nodes that are further away from the central node, see Fig. B.3(a), which indicates that the method performs similarly for the central nodes and the nodes which are on the edges of the graph. Fig. B.3(b) displays the error with respect to the average distance to five closest neighbors. It shows the advantage of the model that being in a cluster or oppositely, being isolated, does not affect the performance of the model.

Appendix C. The effectiveness of the multi-window approach

In the study regarding the effectiveness of multi-window approach over multi-head we analyze the dynamical adjacency and temporal attention coefficients obtained from the two models in question during a cloudy day in Eich, Switzerland. This node is selected due to its central position, marked with turquoise color in Fig. 7. For illustration purposes, the predictions for 6 h ahead made by GCLSTM are also shown in Fig. 5 and compared to temporal-spatial graph attention model. The analysis was made on 30th of August, which was the cloudy day, at 7 a.m. for the next 6 h ahead prediction. We choose this time since it is difficult to make forecast in the morning when the most recent past data is limited to only one hour of data and then followed by night values.

The temporal adjacencies of TSM-GAT and TS-multi-head-GAT for this prediction are shown in Fig. 6. TSM-GAT model has the highest attention in the 15th, then 6th and 7th overlapping window, whereas multi-head model has the highest attention on the 10th, and then 11th and 15th overlapping window. This means that TSM-GAT pays the most attention to the observations in the morning of the current day and then on the last 4 h of non-zero PV production in the previous day. This could be interpreted as looking at the past values during the cloudy morning and the last values before sunset of the previous day. Since the previous day was much sunnier it would not make sense for the algorithm to focus on other parts of the sequence. On the other hand, multi-head approach pays the most attention to the 10th and 11th window, which is during the night and then on the 15th window. It has the highest attention values for the night windows when production is zero, which should not be a relevant part of the input data.

The next part of the study considers the dynamical spatial adjacency. We chose 15th window of the spatial attention for analysis, since both the multi-head and the multi-window architectures have high attention coefficients in the last temporal window. For both models we have chosen the arbitrary value 0.0035 as a threshold, such that nodes with coefficients above the threshold are considered important when making the prediction. Fig. 7 shows the map of important nodes represented with different colors based on the window/head they belong to, when making the prediction for the chosen day. The first difference between the models is that in the multi-window approach the model is focusing on the closer nodes when making prediction for up to 2 h ahead and it is focusing on further away nodes when making a prediction for 4-6 h ahead. Intuitively, this represents a clear advantage compared to multi-head attention, since in the multi-head attention all heads give the highest focus on the similar small neighborhoods. Furthermore, in this example the multi-window approach in the first window, coefficients are above the threshold for the self-attention, which is reasonable since for the short-term forecasts the persistence model gives the best results. Therefore, it could be interpreted as paying the most attention to its own past data when making forecasts up to two hours ahead. However, in the multi-head approach, none of these three attention heads have high self-attention coefficient values in the adjacency. Furthermore, in the multi-head approach the first and the last heads share one out of two nodes with highest attention. In the multi-window approach all three windows have different nodes with highest coefficient, meaning that in each window different nodes are selected as the most important.

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